TAHSIN KHAN

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INTRODUCTION

Robotics engineer at **ERTS Lab**, IIT Bombay, focused on autonomy for ground and aerial robots. My recent work at ERTS Lab includes **cooperative mulit-robot transport, LiDAR and visual SLAM** on UAVs, **monocular autonomy** for Ackermann vehicles, and **remote robotics education platforms**. I have experience in **state estimation**, **SLAM**, **motion planning**, **optimisation-based control**, and **imitation learning** for manipulation. Broadly, my research interests are in **multi-robot systems**, **hybrid model-based and learned approaches** for robot autonomy, with a further interest in **soft robotics and biomimetics**.

EDUCATION

National Institute of Technology Karnataka — Surathkal, India

Bachelor of Technology in Mechanical Engineering — *Dec 2021 – April 2025*

(with Minors in Electronics and Communication Engineering)

CGPA *(Major)*: **8.22/10.00** CGPA *(Minor)*: **8.20/10.00**

PUBLICATIONS

1. **T. Khan** and M. J. Thomas. "Soft autonomous mobile manipulators in agricultural automation: a review." *Industrial Robot*, Vol. **53**, No. 2, pp. 376–398, 2026. doi:10.1108/IR-05-2025-0162

RESEARCH EXPERIENCE

Grasp-Anchored Cooperative Transport — Feb 2026 - Present

Advisors: Prof. Shivaram Kalyanakrishnan, Prof. Harshad Khadilkar, Prof. Nived Chebrolu — *IIT Bombay, India*

- Ongoing work on cooperative transport of a rigid payload by **two custom nonholonomic mobile manipulators**, using **onboard vision and decentralised compute only**: no motion capture, no shared map and **no force/torque sensing** at the grasp.
- Posed planning and control entirely in the **payload frame**, so transport depends only on the **relative pose between the robots**; global localisation reduces to coarse textbf landmark-based estimation of where the payload and bases sit within the workspace.
- Recovering relative pose through the **kinematic chain of the shared grasp** rather than odometry, making relative orientation immune to wheel slip and map drift; a **an EKF on SE(3)** estimates each grasp transform with χ^2 innovation gating, running at **50 Hz** on a Jetson Orin from **30 Hz** wrist camera detections.
- Developed and tested a **14-DoF whole-body QP controller in MuJoCo simulation** before deploying it on hardware, incorporating the closed-chain constraint as a soft cost weighted by live estimator covariance to relax coupling under uncertainty while maintaining solver feasibility during transport.

Embedded Real Time Systems (ERTS) Lab, e-Yantra — Aug 2025 - Present

Senior Project Technical Assistant | Prof. Kavi Arya, Prof. Shivaram Kalyanakrishnan — *IIT Bombay, India*

- Researched and developed **eYRC 2026-27**'s new competition theme, **Niti Vahan**, an autonomous car operating under city road rules, selected by **1500 teams** nationally; worked on **electronics**, **perception** and **control** algorithms for an in-house developed miniature Ackermann vehicle.
- Monocular pipeline on a Raspberry Pi with a single USB camera: **lane detection** via bird's eye view warping, colour segmentation and polynomial fitting under **PID** steering control, with obstacle detection, **lane switching** and **ArUco based docking** for parking; teams build against a matched **CoppeliaSim digital twin** in Stage 1 before moving to physical hardware in Stage 2.
- Developed the **KrishiCobot** theme (eYRC 2025-26) for **649 teams** and **2436 participants**, delivering **475 hours** of remote access to a **UR5** arm and a mobile robot on a mock greenhouse; part of ongoing research on **scalable, low-cost remote laboratories** for robotics education, with a paper in preparation.
- Mentored **three eYSIP summer research projects** and **five interns** in 2026, and delivered faculty training and outreach workshops reaching **120+ participants**.

Visual SLAM for UAVs (eYSIP 2025) May 2025 - Jul 2025
Research Intern | e-Yantra, ERTS Lab *IIT Bombay, India*

- Built a complete visual SLAM **front end and back end from scratch** using only Open3D and OpenCV, after judging ORB-SLAM3 and RTAB-Map impractical to adapt and finding no end-to-end 3D VSLAM plus navigation stack comparable to Nav2.
- Implemented **ICP** registration, keyframe selection and incremental map building, producing consistent maps of indoor environments.
- Added local and global **loop closure** with **pose graph optimisation**, and relocalisation using a **bag of words** place recognition front end refined by ICP.
- Validated handheld on a **RealSense D455** with an **Orin NX**; designed an extension to tightly coupled **VIO** by adding IMU preintegration factors to the pose graph.

Soft Autonomous Mobile Manipulators in Agriculture: A Review Sep 2024 - Apr 2025
Undergraduate Research | Prof. Mervin Joe Thomas *NITK Surathkal, India*

- Conducted a systematic review of **soft mobile manipulators** for agricultural automation, published in **Industrial Robot** (Vol. 53, No. 2, 2026).
- Compared **soft gripping and actuation** technologies across actuator families, assessing compliance, payload and crop damage trade offs for harvesting tasks.
- Reviewed classical and state of the art **localisation and path planning** methods for mobile bases operating in unstructured field environments.
- Analysed the **mobile manipulator design space**, identifying where gripper compliance relaxes precision requirements on the base and arm.

Autonomous Robots and Multi-Systems (ARMS) Lab (eYSIP 2024) May 2024 - Jul 2024
Research Intern | Prof. Arpita Sinha *IIT Bombay, India*

- Built a **ROS 2 controlled 4WD Ackermann** vehicle for field exploration, from low level motor and steering control through to autonomous navigation on the physical platform.
- Fused IMU with Vicon velocity estimates through a **Kalman filter** for odometry, ran **LiDAR SLAM** with **AMCL** localisation in Nav2, and implemented a **pure pursuit** planner for waypoint navigation.

PROJECTS AND COMPETITIONS

Imitation Learning for Arm Manipulation (eYSIP 2026) May 2026 - Jul 2026
Mentor and Co-Mentor, 4 interns | SO-101, UR5, LeRobot, ACT, SmolVLA, Agilex Pika Sense *IIT Bombay, India*

- Defined and supervised **two parallel eYSIP projects** evaluating **imitation learning policies** on real hardware, mentoring 3 interns on a bimanual **SO-101** setup and co-mentoring 1 intern on a **UR5** arm.
- Guided benchmarking of **ACT** and **SmolVLA** across bimanual pick and place, **cube stacking**, precision insertion and single-arm grasping, overseeing **dataset collection**, policy training and an ablation replacing an RGB wrist camera with a **RealSense D405**, requiring modifications to the ACT codebase.
- Supervised development of a **UR5 teleoperation and data pipeline** using **Pika Sense**, with joint states and camera images recorded to **ROS bags** and converted into LeRobot format, enabling imitation learning on a platform not natively supported by **LeRobot**.

3D LiDAR SLAM on a UAV (eYSIP 2026) May 2026 - Jul 2026
Mentor, 1 intern | Holybro X500, Airy LiDAR, Jetson Orin Nano *IIT Bombay, India*

- Mentored an intern in benchmarking **KISS-ICP, LIO-SAM and FAST-LIO** and developing a custom FAST-LIO back end with **keyframe management, pose graph optimisation and loop closure**, validated through mapping experiments at the **KReSIT building** at IIT Bombay.
- Guided UAV hardware integration, including the design of a **power distribution board** and sensor and compute mounts under tight payload constraints, and troubleshooting of map drift associated with **LiDAR–IMU extrinsic calibration errors** and thermal throttling.
- Supervised evaluation of **LIO-SAM** on manual UAV flights, generating 3D maps, with autonomous navigation remaining ongoing.

LeHome Challenge, IEEE ICRA 2026: Bimanual Cloth Folding Jan 2026 - Mar 2026

*Rank **32 of 62** teams internationally | Advisor: Prof. Nived Chebrolu* *IIT Bombay, India*

- Simulation round in **Isaac Sim**: fold any of **four garment categories**, spawned at random, with a bimanual SO-101 setup.
- Compared **four specialist SmolVLA policies** routed by a **YOLO** garment classifier against a **single generalist policy**; both converged to comparable performance, **55% success** locally and **49%** on the final server evaluation.
- Also evaluated **ACT, XVLA, diffusion policy and** π_0, including a SmolVLA and XVLA hybrid for the specialist route.

MuJoCo-ROS2 Quadruped Robot Navigation System Apr 2025 - May 2025

ROS2 Humble, MuJoCo, Python

- Created a **ROS2 bridge** enabling communication between **MuJoCo** and **RViz** for robot visualization, with *2D* and *3D* **IK solvers** calculating *joint angles* from foot positions for **trajectory generation**.
- Implemented a **trotting gait** based quadruped **locomotion** with configurable *stride parameters*, and **waypoint navigation** using *cubic spline* interpolation of RViz interactive markers with heading control.

NITK Autonomous Campus Shuttle Jan 2025 - Dec 2025

ROS 2, Autoware, LIO-SAM | Centre of System Design, NITK *NITK Surathkal, India*

- Led the **software side** of an **L5 autonomous shuttle** built on a **Mahindra Reva** electric chassis, while the wider team handled drive by wire steering and throttle modifications.
- Selected the sensor suite (**3D LiDAR, IMU, cameras**) and set up **Autoware** as the core stack with ROS 2 as middleware.
- Chose and integrated the SLAM stack, deploying **LIO-SAM** on the vehicle, mapping a substantial portion of the NITK campus and demonstrating **point to point autonomous navigation**.

DD Robocon 2023: ABU Robocon National Qualifier Jul 2023

*Mechanical Subsystem | CSD Robocon NITK, top **50 teams** nationally* *Delhi, India*

- Designed **chassis and mechanisms** for two semi autonomous robots as part of NITK's **first ever qualification** for the national stage, including **catapult** and **dual roller** shooting mechanisms for ring throwing onto poles up to **12 metres** away.
- Iterated through **rapid prototyping** with 3D printing and laser cutting, and performed **FEA** on load-bearing members to size the structure against impact loading during launch.

TECHNICAL SKILLS

Programming: Python, C++, Embedded C

Robotics Stacks: ROS, ROS 2, Nav2, MoveIt, Autoware, LeRobot, Open3D, OpenCV, OSQP

Simulation: Gazebo, MuJoCo, Isaac Sim, CoppeliaSim

Hardware: NVIDIA Jetson (Orin Nano, Orin NX), Raspberry Pi, ESP32, Pixhawk, UR5, Kinova, SO-101, CAD (Fusion 360, Onshape).

Relevant Coursework: Basics of Robotics and Embedded Systems, Control Engineering, Signals and Systems, Mechatronic Systems, Microsystem Technology, Digital Electronics, Deep Learning Specialization (MOOC)

TEACHING, OUTREACH AND ACHIEVEMENTS

- Delivered two day hands on **embedded systems Faculty Development Programmes** on the AlphaBot v2 platform for approximately **80 college professors** in Bengaluru and Mysore, representing IIT Bombay's ERTS Lab.
- Conducted a two day **ROS workshop for 40+ participants** as part of e-Yantra's *Women in Robotics* initiative.
- Ranked **32 of 62 teams internationally** in the LeHome Challenge at IEEE ICRA 2026.
- Selected for the e-Yantra Summer Internship Programme at IIT Bombay, among the **top 80 applicants** nationally.
- **Electronics SIG Head**, SAE NITK (2024-25): led year long student projects in autonomous vehicles and robotics under the SAE Autonomous Mobility theme.